

Reliability of a Flow Network Subject to Budget Constraints

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Abstract—The system capacity of a deterministic flow network is the maximum flow from the source to the destination. In a single-commodity stochastic-flow network (branches all have several possible capacities, and may fail), the system reliability, the probability that the maximum flow is larger than or equal to a given demand, is an important performance index to measure the quality level of a network. In a two-commodity stochastic-flow network, different types of commodities are transmitted through the same network simultaneously, and compete for the capacities. We concentrate on the reliability problem for such a network subject to the budget constraint. This paper defines firstly the system capacity as a pattern. We propose a performance index, the probability that the system capacity is less than or equal to a given pattern subject to the budget constraint, to evaluate the system performance. A simple algorithm based on minimal cuts is proposed to generate all maximal vectors meeting the demand and budget constraints. The performance index can then be computed in terms of all such maximal vectors.

Index Terms—Budget, capacity vectors, minimal cuts, performance index, reliability, two-commodity.

ACRONYMS¹

MP	minimal path
MC	minimal cut
SFN	stochastic-flow network
SSFN	single-commodity stochastic-flow network
TSFN	two-commodity stochastic-flow network

NOMENCLATURE

G	(A, N, M) : a SFN
A, n	set of branches, number of branches
N	set of nodes
a_i	branch $\#i, i = 1, 2, \dots, n$
s, t	source node, destination node
x_i	current capacity of $a_i, i = 1, 2, \dots, n$
X	$(x_1, x_2, \dots, x_n)$: capacity vector
M_i	maximal capacity of $a_i, i = 1, 2, \dots, n$
M	$(M_1, M_2, \dots, M_n)$: maximal capacity vector
c_i	transmission cost per unit capacity of $a_i, i = 1, 2, \dots, n$

$C(X)$	$\sum_{i=1}^n c_i x_i$: total cost under X
B	budget
d	demand at t in a SSFN
d_i	required quantity of commodity i at t in a TSFN, $i = 1, 2$
α_i	consumed quantity of capacity on each branch per unit of commodity $i, i = 1, 2$
m	number of MC
K_r	MC $\#r, r = 1, 2, \dots, m$
n_r	number of branches in K_r
f_i	flow of commodity 1 through $a_i, i = 1, 2, \dots, n$
h_i	flow of commodity 2 through $a_i, i = 1, 2, \dots, n$
$V(X)$	system capacity under X in a TSFN
$W(X)$	system capacity under X in a SSFN i.e., the maximum value of flow from s to t
$U_{d_1, d_2, B}$	probability that the system capacity is less than or equal to (d_1, d_2) subject to budget constraint B .
Ω	$\{X X \text{ supports at most } (d_1, d_2) \text{ and meeting } C(X) \leq B\}$: the set of candidates of (d_1, d_2, B) -MC
$\Omega_{\max}$	$\{X X \text{ is a maximal vector in } \Omega\}$
w	number of (d_1, d_2, B) -MCs
B_i	$\{X X \leq X_i\}, i = 1, 2, \dots, w$
$[x]$	largest integer such that $[x] \leq x$

I. INTRODUCTION

FROM the point of view of quality management and decision making, it is an important issue to define the system capacity, and then to develop some indexes in order to evaluate the system performance. We can treat many real-life systems such as computer systems, telecommunication systems, urban traffic systems, and logistics systems, as flow networks. Traditionally, under the condition that the branch capacity is deterministic, the system capacity is the maximum value of the flow from s to t . In many real networks, the branch capacity should be stochastic due to the fact that the branch may be in failure, maintenance, etc. Such a network is called a Stochastic-Flow Network (SFN) throughout this paper. Hence, its system capacity is not a fixed number. For instance, a computer system can be modeled as a SFN in which branches stand for transmission lines, and nodes stand for switches. Virtually, each transmission line consists of several physical lines, and each physical line has successful and failed states. That implies a transmission line has several states in which state k means that k physical lines are successful. Hence, the capacity of each branch has several values.

Manuscript received October 9, 2005. This work was supported in part by the National Science Council, Taiwan, R.O.C., under Grant NSC 93-2213-E-238-002. Associate Editor: C. Smith.

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Digital Object Identifier 10.1109/TR.2006.884594

¹The singular & plural of an acronym are always spelled the same.

For a single-commodity SFN (SSFN), several authors [9], [12], [20], [21] evaluated the probability that the system capacity is larger than or equal to the demand d (integer) without a budget constraint. Such a probability, which is in general called the system reliability, is a performance index useful to measure the quality level of SSFN. They evaluated the system reliability by generating all d -MP in terms of MP, where a MP is a set of branches whose proper subsets are no longer paths. A d -MP is a minimal capacity vector which enables the system to transmit d flow from s to t . Lin [10] extended the reliability problem to include the budget constraint in terms of MP. The max-flow min-cut Theorem [5] says that the maximum value of the flow from s to t equals the minimum capacity among all MC. A MC is a set of branches whose proper subsets are no longer cuts. Hence, Jane *et al.* [7], Lin [11], and Yeh [21] used MC to evaluate another performance index, the probability that the system capacity is less than or equal to d , by generating all d -MC. A d -MC is a maximal capacity vector which enables the system to transmit exactly d flow from s to t . Such a performance index is in general called the system unreliability in SSFN.

However, many SFN allow multiple types of commodity to be transmitted from s to t simultaneously, especially in the case that different commodities consume the branch's capacity differently. For instance, a broadband computer network is a typical TSFN in which two-commodity (audio files denote commodity 1, video files denote commodity 2) share the bandwidth (capacity) simultaneously. Lin [16] extended the reliability problem from SSFN to TSFN without a budget constraint. This paper concentrates on the reliability problem for a TSFN subject to the budget constraint. We will define the system capacity as a 2-tuple vector in terms of MC, and then propose a performance index, the probability that the system capacity is less than or equal to a given pattern subject to the budget constraint, in Section II. An algorithm is proposed in Section III to generate all maximal vectors meeting the demand & budget constraints in order to evaluate the performance index. An international trade example is shown in Section IV to illustrate the proposed approach, and how the performance index may be calculated. Time complexity, and storage complexity of the proposed algorithm are both analysed in Section V.

Assumptions:

- 1) The current capacity x_i is an integer-valued random variable which takes the value from $\{0, 1, 2, \dots, M_i\}$ according to a given probability distribution.
- 2) Without loss of generality for α_i , $\alpha_2 \geq \alpha_1 = 1$.
- 3) Two types of commodities are transmitted from s to t .
- 4) The capacities of different branches are statistically independent.
- 5) The flow in G must satisfy the flow-conservation law [5].

II. PROBLEM FORMULATION

A. Flow Assignment in the TSFN

Vector comparisons are made as follows:

$$\begin{aligned} (y_1, y_2, \dots, y_n) &\geq (x_1, x_2, \dots, x_n): y_i \geq x_i \text{ for each } i = 1, 2, \dots, n \\ (y_1, y_2, \dots, y_n) &> (x_1, x_2, \dots, x_n): (y_1, y_2, \dots, y_n) \geq (x_1, x_2, \dots, x_n) \text{ and } y_i > x_i \text{ for at least one } i \end{aligned}$$

$$\begin{aligned} (d'_1, d'_2) &\geq (d_1, d_2): d'_1 \geq d_1 \text{ and } d'_2 \geq d_2 \\ (d'_1, d'_2) &> (d_1, d_2): (d'_1, d'_2) \geq (d_1, d_2) \text{ and } d'_i \geq d_i \text{ for at least one } i \end{aligned}$$

With respect to each MC, $K_r = \{a_{r1}, a_{r2}, \dots, a_{rn_r}\}$, the vector $(f_{r1}, f_{r2}, \dots, f_{rn_r}, h_{r1}, h_{r2}, \dots, h_{rn_r})$ is feasible under the capacity vector $X = (x_1, x_2, \dots, x_n)$ if

$$\alpha_1 f_{rj} + \alpha_2 h_{rj} \leq x_{rj} \quad \text{for } j = 1, 2, \dots, n_r \quad (1)$$

where $(\alpha_1 f_{rj} + \alpha_2 h_{rj})$ is the consumed quantity of capacity on a_{rj} by the flows f_{rj} & h_{rj} . Such a vector is called a flow assignment (with respect to K_r).

Under the capacity vector X , the MC K_r is said to support the demand (d_1, d_2) if there exists an $(f_{r1}, f_{r2}, \dots, f_{rn_r}, h_{r1}, h_{r2}, \dots, h_{rn_r})$ feasible under X such that

$$\sum_{j=1}^{n_r} f_{rj} = d_1 \text{ \& \> } \sum_{j=1}^{n_r} h_{rj} = d_2. \quad (2)$$

Under X , K_r is said to support at most (d_1, d_2) (i.e., K_r supports (d_1, d_2) but cannot support any (d'_1, d'_2) with $(d'_1, d'_2) > (d_1, d_2)$) if K_r supports (d_1, d_2) , and there exists no $(f_{r1}, f_{r2}, \dots, f_{rn_r}, h_{r1}, h_{r2}, \dots, h_{rn_r})$ feasible under X such that $\sum_{j=1}^{n_r} f_{rj} = d_1 + 1$ & $\sum_{j=1}^{n_r} h_{rj} = d_2$. Moreover, the capacity vector X is said to support (d_1, d_2) if, under X , all MC support (d_1, d_2) . Then X is said to support at most (d_1, d_2) if, under X , all MC support (d_1, d_2) , and at least one MC supports at most (d_1, d_2) . In a TSFN, the system capacity $V(X)$ is defined to be (d_1, d_2) if X supports at most (d_1, d_2) .

Without budget constraint, two performance indexes, $\Pr\{X|V(X) \geq (d_1, d_2)\}$, and $\Pr\{X|V(X) \leq (d_1, d_2)\}$, can be adopted to evaluate the performance for a TSFN; the former is the probability that the system capacity is larger than or equal to (d_1, d_2) , and the latter is the probability that the system capacity is less than or equal to (d_1, d_2) . Lin [12], [14] had proposed algorithms to evaluate $\Pr\{X|V(X) \geq (d_1, d_2)\}$ in terms of MP. This paper concentrates on those X which satisfy $V(X) \leq (d_1, d_2)$, and the constraint

$$C(X) \leq B. \quad (3)$$

That is, we evaluate the performance index $U_{d_1, d_2, B}$, which equals $\Pr\{X|V(X) \leq (d_1, d_2) \text{ \& \> } C(X) \leq B\}$.

B. Definition of (d_1, d_2, B) -MC

The capacity vector X is a (d_1, d_2, B) -MC if

- i) $V(X) = (d_1, d_2)$,
- ii) $C(X) \leq B$, and
- iii) $V(Y) > (d_1, d_2)$ or $C(Y) > B$

for any capacity vector Y with $Y > X$. The set of (d_1, d_2, B) -MC is the set of maximal vectors in $\{X|V(X) = (d_1, d_2) \text{ \& \> } C(X) \leq B\}$, and also the set of maximal ones in $\{X|V(X) \leq (d_1, d_2) \text{ \& \> } C(X) \leq B\}$. We have the following result.

Lemma 1: If X is a (d_1, d_2, B) -MC, then $V(Y) \leq (d_1, d_2)$ & $C(Y) \leq B$ for each capacity vector Y with $Y < X$.

Thus, the performance index $\Pr\{X|V(X) \leq (d_1, d_2) \text{ \& \> } C(X) \leq B\}$ equals $\Pr\{X|X \leq$

X_i for a (d_1, d_2, B) – MC X_i . The following Theorem shows a necessary condition for a capacity vector X to be a (d_1, d_2, B) -MC.

Theorem 1: For each (d_1, d_2, B) -MC, there exists a $K_r = \{a_{r1}, a_{r2}, \dots, a_{rn_r}\}$, and an $(f_{r1}, f_{r2}, \dots, f_{rn_r}, h_{r1}, h_{r2}, \dots, h_{rn_r})$, such that

$$\sum_{j=1}^{n_r} f_{rj} = d_1 \ \& \ \sum_{j=1}^{n_r} h_{rj} = d_2 \quad (4)$$

and

$$\begin{cases} x_{rj} = \alpha_1 f_{rj} + \alpha_2 h_{rj} & \text{for } j = 1, 2, \dots, n_r \\ x_i = M_i & \text{for } a_i \notin K_r \end{cases} \quad (5)$$

With respect to each K_r , we generate an X via (5) for each flow assignment $(f_{r1}, f_{r2}, \dots, f_{rn_r}, h_{r1}, h_{r2}, \dots, h_{rn_r})$ satisfying (4). For each $a_{rj} \in K_r$, x_{rj} supports at most (f_{rj}, h_{rj}) . The branches in K_r play the roles as the parallel bridges in transportation. Hence, K_r supports at most $(\sum_{j=1}^{n_r} f_{rj}, \sum_{j=1}^{n_r} h_{rj}) = (d_1, d_2)$. That is, such an X supporting at most (d_1, d_2) , and meeting $C(X) \leq B$, is a candidate of (d_1, d_2, B) -MC. Theorem 1 implies that we have the following result.

Lemma 2: Each (d_1, d_2, B) -MC is a candidate of (d_1, d_2, B) -MC.

Theorem 2 further shows that $\Omega_{\max}$ is the set of (d_1, d_2, B) -MC.

Theorem 2: $\Omega_{\max} = \{(d_1, d_2, B)\text{-MC}\}$.

C. Evaluate $U_{d_1, d_2, B}$

If $\Omega_{\max} = \{X_1, X_2, \dots, X_w\}$, the performance index $U_{d_1, d_2, B}$ can be computed in terms of all (d_1, d_2, B) -MC by

$$\begin{aligned} \Pr\{X|V(X) \leq (d_1, d_2) \ \& \ C(X) \leq B\} \\ &= \Pr\{X|X \leq X_i \text{ for a } (d_1, d_2, B) \text{ - MC } X_i\} \\ &= \Pr\{B_1 \cup B_2 \cup \dots \cup B_w\}. \end{aligned} \quad (6)$$

This value can be calculated by applying methods such as the inclusion-exclusion method [6], [11]–[16], disjoint subsets [19], and state-space decomposition [2], [3], [7]–[10], [17]. Note that $\Pr\{Y \leq X\} = \Pr\{y_1 \leq x_1\} \times \Pr\{y_2 \leq x_2\} \times \dots \times \Pr\{y_n \leq x_n\}$ by Assumption 4, if $Y = (y_1, y_2, \dots, y_n)$. In particular, the inclusion-exclusion method says that $\Pr\{B_1 \cup B_2 \cup \dots \cup B_w\} = \sum_i \Pr\{B_i\} - (-1)^2 \sum_{i < j} \Pr\{B_i \cap B_j\} - (-1)^3 \sum_{i < j < k} \Pr\{B_i \cap B_j \cap B_k\} - \dots - (-1)^w \Pr\{B_1 \cap B_2 \cap \dots \cap B_w\}$.

III. ALGORITHM

As in those approaches taken in [7], [11], [19], [21], we suppose that all MC have been pre-computed. All MC can be efficiently derived from the algorithms discussed in [1], [4], [18]. All (d_1, d_2, B) -MC can be determined by the following steps.

Step 1. With respect to each MC $K_r = \{a_{r1}, a_{r2}, \dots, a_{rn_r}\}$, perform the following steps.

1.1 Find all flow assignments $(f_{r1}, f_{r2}, \dots, f_{rn_r}, h_{r1}, h_{r2}, \dots, h_{rn_r})$ satisfying the capacity, and demand constraints of

$$\alpha_1 f_{rj} + \alpha_2 h_{rj} \leq M_{rj} \quad \text{for } j = 1, 2, \dots, n_r, \quad (7)$$

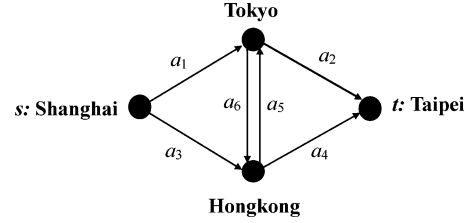


Fig. 1. A benchmark network.

$$\sum_{j=1}^{n_r} f_{rj} = d_1 \ \& \ \sum_{j=1}^{n_r} h_{rj} = d_2. \quad (8)$$

1.2 Transform each flow assignment in step 1.1 into $X = (x_1, x_2, \dots, x_n)$ via

$$\begin{cases} x_{rj} = \alpha_1 f_{rj} + \alpha_2 h_{rj} & \text{for } j = 1, 2, \dots, n_r \\ x_i = M_i & \text{for } a_i \notin K_r. \end{cases} \quad (9)$$

1.3 Store X into Ω if

$$\sum_{i=1}^n c_i x_i \leq B. \quad (10)$$

Step 2. (Obtain $\Omega_{\max}$) Suppose $\Omega = \{X_1, X_2, \dots, X_w\}$. Remove those non-maximal ones from Ω to obtain $\Omega_{\max}$.

2.1) $I \leftarrow \phi, \Omega_{\max} \leftarrow \phi$.

2.2) For $i \leftarrow 1$ to w with $i \notin I$.

2.3) For $j \leftarrow i + 1$ to w with $j \notin I$.

2.4) If $X_j \geq X_i$, then $X_i \notin \Omega_{\max}, I \leftarrow I \cup \{i\}$, and Goto 2.7).

Else If $X_i > X_j$, then $I \leftarrow I \cup \{j\}$.

2.5) $j \leftarrow j + 1$.

2.6) $\Omega_{\max} \leftarrow \Omega_{\max} \cup X_i$.

2.7) $i \leftarrow i + 1$.

2.8) END.

IV. A NUMERICAL EXAMPLE

Fig. 1 shows an international trade network. According to the law of Taiwan, no direct route from s (Shanghai, China) to t (Taipei, Taiwan) is permitted. Hence, each route from s should pass Hong Kong or Tokyo before arriving t . The supplier would like to transmit 15-inch LCD monitors (commodity 1), and 17-inch LCD monitors (commodity 2) from Shanghai to Taipei. All commodities are loaded into the same type of container. The capacity of each arc is stochastic due to the fact that either containers or traffic tools through each arc may be in maintenance, reserved by other suppliers, or in other conditions. One unit of commodity means 120 homogeneous commodities. The supplier wants to have its 2 units of 15-inch LCD monitors, and 2 units of 17-inch LCD monitors (i.e., $d_1 = 2$, and $d_2 = 2$) to be transported from s to t . The sizes of each container, 15-inch LCD monitor, and 17-inch LCD monitor, are $591 \times 230 \times 220$ (length $\times$ breadth $\times$ high) cm^3 , $92 \times 52 \times 38 \text{ cm}^3$, and $125 \times 70 \times 43 \text{ cm}^3$, respectively. Hence,

TABLE I
THE DATA OF BRANCHES FOR FIG. 1

Branch	Capacity	Cumulative Probability	Cost **	α_1	α_2	Branch	Capacity	Cumulative Probability	Cost	α_1	α_2
a_1	4	1.00	250			a_4	4	1.00	150		
	3*	0.50					3	0.50			
	2	0.30					2	0.30			
	1	0.20					1	0.20			
	0	0.10					0	0.10			
a_2	3	1.00	250	1	2	a_5	3	1.00	250	1	2
	2	0.40					2	0.40			
	1	0.20					1	0.20			
	0	0.10					0	0.10			
a_3	3	1.00	300			a_6	3	1.00	150		
	2	0.40					2	0.40			
	1	0.20					1	0.20			
	0	0.10					0	0.10			

* the cumulative probability: $\Pr\{x_1 \leq 3\} = 0.5$

** US dollar

one unit of commodity 1 (resp. 2) consumes 1 (resp. 2) container, i.e., $\alpha_1 = 1$ (resp. $\alpha_2 = 2$). The cost c_i through each branch a_i is counted by the number of containers.

The arc data are given in Table I. In this example, $(M_1, M_2, M_3, M_4, M_5, M_6) = (4, 3, 3, 4, 3, 3)$, and there are 4 MC: $K_1 = \{a_1, a_3\}$, $K_2 = \{a_2, a_4\}$, $K_3 = \{a_1, a_4, a_5\}$ & $K_4 = \{a_2, a_3, a_6\}$. The manager would like to know the probability that the system capacity is less than or equal to (2, 2), subject to the budget $B = 4400$ US dollars. We first generate all (2, 2, 4400)-MC as follows.

Step 1. With respect to $K_1 = \{a_1, a_3\}$, perform the following steps.

1.1 Find all flow assignments (f_1, f_3, h_1, h_3) satisfying the constraints

$$\begin{cases} f_1 + 2h_1 \leq 4 \\ f_3 + 2h_3 \leq 3 \\ f_1 + f_3 = 2 \\ h_1 + h_3 = 2 \end{cases}$$

We obtain three (f_1, f_3, h_1, h_3) : (1, 1, 1, 1), (2, 0, 1, 1), and (0, 2, 2, 0).

1.2 Transform each (f_1, f_3, h_1, h_3) into $X = (x_1, x_2, \dots, x_6)$ via

$$\begin{cases} x_1 = f_1 + 2h_1, x_3 = f_3 + 2h_3 \\ x_2 = M_2, x_4 = M_4, x_5 = M_5, x_6 = M_6. \end{cases}$$

Two different X are generated: $X_1 = (3, 3, 3, 4, 3, 3)$, and $X_2 = (4, 2, 3, 4, 3, 3)$.

1.3 Both X satisfy the budget constraint as

$$C(X_1) = 4200 < 4400, \text{ and } C(X_2) = 4150 < 4400.$$

So $\Omega = \{X_1, X_2\}$.

Step 1. With respect to $K_2 = \{a_2, a_4\}$.

1.1 Obtain all (f_2, f_4, h_2, h_4) of the constraints

$$\begin{cases} f_2 + 2h_2 \leq 3 \\ f_4 + 2h_4 \leq 4 \\ f_2 + f_4 = 2 \\ h_2 + h_4 = 2 \end{cases}$$

We obtain three (f_2, f_4, h_2, h_4) : (1, 1, 1, 1), (2, 0, 0, 2), and (0, 2, 1, 1).

1.2 Transform each (f_2, f_4, h_2, h_4) into $X = (x_1, x_2, \dots, x_6)$ via $x_2 = f_2 + 2h_2$, $x_4 = f_4 + 2h_4$, and $x_i = M_i$ for others. Then two different X are generated: $X_3 = (4, 3, 3, 3, 3, 3)$, and $X_4 = (4, 3, 2, 4, 3, 3)$.

1.3 Both $C(X_3) = 4300$, and $C(X_4) = 4200$ do not exceed the budget 4400, so $\Omega = \{X_1, X_2, X_3, X_4\}$.

Repeating the procedures for K_3 to K_4 , in sum we obtain 29 candidates, and the result is shown in Table II.

Step 2. $\Omega = \{X_1, X_2, \dots, X_{29}\}$.

2.1) $I \leftarrow \phi$, $\Omega_{\max} \leftarrow \phi$

2.2) $i \leftarrow 1$

2.3) $j \leftarrow 2$

2.4) $X_2 = (4, 2, 3, 4, 3, 3) \not\geq X_1 = (3, 3, 3, 4, 3, 3)$, and $X_1 \not\geq X_2$. $I = \phi$.

2.3) $j \leftarrow 3$

2.4) $X_3 \not\geq X_1$, and $X_1 \not\geq X_3$. $I \leftarrow \phi$.

2.3) $j \leftarrow 4$

2.4) $X_4 \not\geq X_1$, and $X_1 \not\geq X_4$. $I \leftarrow \phi$.

2.3) $j \leftarrow 5$

2.4) $X_1 = (3, 3, 3, 4, 3, 3) > X_5 = (2, 3, 3, 4, 0, 3)$. $I = \{5\}$.

2.3) $j \leftarrow 6$

2.4) $X_1 = (3, 3, 3, 4, 3, 3) > X_6 = (2, 3, 3, 2, 2, 3)$. $I = \{5, 6\}$.

:

2.6) X_1 is a (2,2,4400)-MC.

2.2) $i \leftarrow 2$

2.3) $j \leftarrow 3$

:

2.8) END.

The result is listed in Table II.

TABLE II
THE RESULT OF THE NUMERICAL EXAMPLE FOR CASE (2,2,4400)

MC	Candidate of (2,2,4400)-MC	$C(X)$	Is a (2,2,4400)-MC	MC	Candidate of (2,2,4400)-MC	$C(X)$	Is a (2,2,4400)-MC
K_1	$X_1 = (3,3,3,4,3,3)$	4200	YES	K_3	$X_5 = (2,3,3,4,0,3)$	3200	NO
	$X_2 = (4,3,2,4,3,3)$	4150	YES		$X_6 = (2,3,3,2,2,3)$	3400	NO
	$X_3 = (4,3,3,3,3,3)$	4300	YES		$X_7 = (4,3,3,2,0,3)$	3400	NO
K_4	$X_4 = (4,2,3,4,3,3)$	4200	YES	K_4	$X_8 = (4,3,3,0,2,3)$	3600	NO
	$X_{20} = (4,2,2,4,3,2)$	3750	NO		$X_9 = (0,3,3,4,2,3)$	3200	NO
	$X_{21} = (4,3,3,4,3,0)$	4000	YES		$X_{10} = (3,3,3,3,0,3)$	3300	NO
	$X_{22} = (4,3,1,4,3,2)$	3700	NO		$X_{11} = (3,3,3,1,2,3)$	3500	NO
	$X_{23} = (4,1,3,4,3,2)$	3800	NO		$X_{12} = (1,3,3,3,2,3)$	3300	NO
	$X_{24} = (4,3,2,4,3,1)$	3850	NO		$X_{13} = (1,3,3,4,1,3)$	3200	NO
	$X_{25} = (4,3,0,4,3,3)$	3550	NO		$X_{14} = (3,3,3,2,1,3)$	3400	NO
	$X_{26} = (4,0,3,4,3,3)$	3700	NO		$X_{15} = (3,3,3,0,3,3)$	3600	NO
	$X_{27} = (4,1,2,4,3,3)$	3650	NO		$X_{16} = (0,3,3,3,3,3)$	3300	NO
	$X_{28} = (4,2,1,4,3,3)$	3600	NO		$X_{17} = (1,3,3,2,3,3)$	3400	NO
	$X_{29} = (4,2,3,4,3,1)$	3900	NO		$X_{18} = (2,3,3,1,3,3)$	3500	NO
					$X_{19} = (2,3,3,3,1,3)$	3300	NO

TABLE III
THE RESULT OF THE NUMERICAL EXAMPLE FOR CASE (2,2,4200)

MC	Candidate of (2,2,4200)-MC	$C(X)$	Is a (2,2,4200)-MC	MC	Candidate of (2,2,4200)-MC	$C(X)$	Is a (2,2,4200)-MC
K_1	$X_1 = (3,3,3,4,3,3)$	4200	YES	K_3	$X_5 = (2,3,3,4,0,3)$	3200	NO
	$X_2 = (4,3,2,4,3,3)$	4150	YES		$X_6 = (2,3,3,2,2,3)$	3400	NO
K_2	$X_3 = (4,3,3,3,3,3)$	4300	NO	K_4	$X_7 = (4,3,3,2,0,3)$	3400	YES
	$X_4 = (4,2,3,4,3,3)$	4200	YES		$X_8 = (4,3,3,0,2,3)$	3600	YES
K_4	$X_{20} = (4,2,2,4,3,2)$	3750	NO	K_4	$X_9 = (0,3,3,4,2,3)$	3200	NO
	$X_{21} = (4,3,3,4,3,0)$	4000	YES		$X_{10} = (3,3,3,3,0,3)$	3300	NO
	$X_{22} = (4,3,1,4,3,2)$	3700	NO		$X_{11} = (3,3,3,1,2,3)$	3500	NO
	$X_{23} = (4,1,3,4,3,2)$	3800	NO		$X_{12} = (1,3,3,3,2,3)$	3300	NO
	$X_{24} = (4,3,2,4,3,1)$	3850	NO		$X_{13} = (1,3,3,4,1,3)$	3200	NO
	$X_{25} = (4,3,0,4,3,3)$	3550	NO		$X_{14} = (3,3,3,2,1,3)$	3400	NO
	$X_{26} = (4,0,3,4,3,3)$	3700	NO		$X_{15} = (3,3,3,0,3,3)$	3600	NO
	$X_{27} = (4,1,2,4,3,3)$	3650	NO		$X_{16} = (0,3,3,3,3,3)$	3300	NO
	$X_{28} = (4,2,1,4,3,3)$	3600	NO		$X_{17} = (1,3,3,2,3,3)$	3400	NO
	$X_{29} = (4,2,3,4,3,1)$	3900	NO		$X_{18} = (2,3,3,1,3,3)$	3500	NO
					$X_{19} = (2,3,3,3,1,3)$	3300	NO

Five potential candidates are (2, 2, 4400)-MC: X_1 , X_2 , X_3 , X_4 , and X_{21} . To compute $U_{2,2,4400} = \Pr\{X|V(X) \leq (2,2) \& C(X) \leq 4400\}$, we let $B_1 = \{X|X \leq X_1\}$, $B_2 = \{X|X \leq X_2\}$, $B_3 = \{X|X \leq X_3\}$, $B_4 = \{X|X \leq X_4\}$, and $B_5 = \{X|X \leq X_{21}\}$. Then $\Pr\{X|V(X) \leq (2,2) \& C(X) \leq 4400\} = \Pr\{B_1 \cup B_2 \cup B_3 \cup B_4 \cup B_5\} = 0.919$ after calculation. In the calculating process,

$$\begin{aligned}
& \Pr\{B_1\} \\
&= \Pr\{X \leq (3,3,3,4,3,3)\} \\
&= \Pr\{x_1 \leq 3\} \times \Pr\{x_2 \leq 3\} \times \Pr\{x_3 \leq 3\} \\
&\quad \times \Pr\{x_4 \leq 4\} \times \Pr\{x_5 \leq 3\} \times \Pr\{x_6 \leq 3\} \\
&= 0.50. \\
& \Pr\{B_1 \cap B_2\} \\
&= \Pr\{(X \leq (3,3,3,4,3,3)) \cap (X \leq (4,2,3,4,3,3))\} \\
&= \Pr\{X \leq (3,2,3,4,3,3)\} \\
&= 0.20.
\end{aligned}$$

Thus, the performance index $U_{2,2,4400}$, the probability that the system capacity is less than or equal to (2,2) subject to the budget $B = 4400$, is 0.919.

For the case that the budget is deduced to be 4200, Table III shows that all (2, 2, 4200)-MC are X_1 , X_2 , X_4 , X_7 , X_8 , and X_{21} . Then the performance index $U_{2,2,4200} = \Pr\{X|V(X) \leq (2,2) \& C(X) \leq 4200\}$ is reduced to be 0.84772 after calculation.

V. COMPUTATIONAL COMPLEXITY ANALYSIS

The number of feasible solutions of (8) is $\binom{n_r+d_1-1}{d_1} \cdot \binom{n_r+d_2-1}{d_2}$. Hence, the number of solutions of constraints (7) & (8) is bounded by $\binom{n_r+d_1-1}{d_1} \cdot \binom{n_r+d_2-1}{d_2}$. Thus, the number of X generated by K_r is bounded by $\binom{n_r+d_1-1}{d_1} \cdot \binom{n_r+d_2-1}{d_2}$. Hence, the number of $X \in \Omega$ is bounded by $\Psi \equiv \sum_{r=1}^m \{\binom{n_r+d_1-1}{d_1} \cdot \binom{n_r+d_2-1}{d_2}\}$. We need at most $O(n\Psi)$ storage space to execute the proposed algorithm.

For MC K_r , each solution of (8) needs $O(n_r)$ time to test all branches whether they satisfy (7) or not. Hence, it takes $O(n_r \binom{n_r+d_1-1}{d_1} \cdot \binom{n_r+d_2-1}{d_2})$ time to obtain all solutions of (7) & (8) in the worst case. It further needs $O(n)$ time to transform each solution in step 1.1 into X via (9), and to check the budget constraint (10). In the worst case, it takes $O(n \binom{n_r+d_1-1}{d_1} \cdot \binom{n_r+d_2-1}{d_2}) = O(n \binom{n_r+d_1-1}{d_1} \cdot \binom{n_r+d_2-1}{d_2}) + O(nr \binom{n_r+d_1-1}{d_1} \cdot \binom{n_r+d_2-1}{d_2})$

TABLE IV
TSFN vs. SSFN

(1,1, B)-MCs in a TSFN	4-MCs in a SSFN [7], [9] – [11], [18] – [20]
(4,3,0,4,3,3), (1,3,3,4,3,3), (3,3,1,4,3,3), (4,1,3,3,3,3), (4,3,3,1,3,3), (4,0,3,4,3,3), (4,1,3,4,3,0), (4,3,1,4,3,0)	(4,3,0,4,3,3), (1,3,3,4,3,3), (3,3,1,4,3,3), (2,3,2,4,3,3)*, (4,1,3,3,3,3), (4,3,3,1,3,3), (4,0,3,4,3,3), (4,2,3,2,3,3)*, (2,3,3,2,0,3)*, (4,1,3,4,3,0), (4,3,1,4,3,0), (4,2,2,4,3,0)*, (4,2,1,4,3,1)*, (4,1,2,4,3,1)*, (4,1,1,4,3,2)*

* not in $\{X|V(X) \leq (1, 1) \text{ \& } C(X) \leq B, B \text{ is unlimited}\}$

time to find all feasible X with respect to K_r . Note that n_r is less than n . Hence, it takes $O(n\Psi)$ time to generate Ω in the worst case. Moreover, it takes $O(n\Psi)$ time to test each element of Ω whether it is maximal in Ω , and $O(n\Psi^2)$ time for all elements of Ω in the worst case. In total, the computational time complexity of the algorithm in the worst case is $O(n\Psi^2) = O(n\Psi) + O(n\Psi^2)$.

VI. CONCLUSIONS AND DISCUSSION

For a TSFN, this paper firstly applies the properties of MC to define the system capacity $V(X)$ as (d_1, d_2) if at most d_1 quantity of commodity 1, and d_2 quantity of commodity 2, can be transmitted under X . An algorithm is proposed to generate all (d_1, d_2, B) -MC, i.e., the maximal ones in the set $\{X|V(X) = (d_1, d_2) \text{ \& } C(X) \leq B\}$. The performance index $U_{d_1, d_2, B}$, the probability that $V(X)$ is larger than or equal to (d_1, d_2) subject to the budget B , is then computed in terms of (d_1, d_2, B) -MC.

It seems that the equation $V(X) = (d_1, d_2)$ implies that $V(X) = (d_1 - k, d_2 + [k\alpha_1/\alpha_2])$ where $k \leq d_1$. We use the example in Section IV to illustrate that this statement only holds in some special cases. The capacity vector $X = (4, 3, 2, 4, 3, 3)$ supports at most (4, 1) whenever $(f_1, f_2, h_1, h_2) = (4, 0, 0, 1)$ with respect to K_1 , and also supports at most $(d_1, d_2) = (2, 2)$ whenever $(f_1, f_3, h_1, h_3) = (2, 0, 1, 1)$ with respect to K_1 . Hence, the system capacity in TSFN is not a unique 2-tuple vector. Conversely, the capacity vector $X = (3, 3, 1, 4, 3, 3)$ supports at most (2, 1) whenever $(f_1, f_3, h_1, h_3) = (1, 1, 1, 0)$ with respect to K_1 , but X cannot support (0, 2).

If the budget B is unlimited (i.e., without budget constraint), set $d = \alpha_1 d_1 + \alpha_2 d_2$, and treat the original problem (in TSFN) as a single-commodity case (in SSFN). Then the approaches [7], [9]–[11], [19]–[21] can be applied to evaluate $\Pr\{X|W(X) \leq d\}$. It seems that the set $\{X|V(X) \leq (d_1, d_2) \text{ \& } C(X) \leq B\}$ is the same as $\{X|W(X) \leq d\}$. We use the network of Fig. 1 to illustrate the difference between them. If $\alpha_2 = 3$, and $(d_1, d_2) = (1, 1)$, then we generate eight $(1, 1, B)$ -MC. Set $d = 1 + 1 \times 3 = 4$, then fifteen 4-MC can be generated. The result concluded in Table IV shows that $\Pr\{X|W(X) \leq 4\}$ is greater than $\Pr\{X|V(X) \leq (1, 1) \text{ \& } C(X) \leq B\}$ because the latter set of X is a proper subset of the former set. This example indicates that the performance evaluation problem in TSFN cannot be simplified to be the one in SSFN.

The proposed algorithm is designed for case $\alpha_1 = 1$. For the more general case that $\alpha_1 \geq 2$, step 1.2 in the proposed algorithm should be modified as follows.

1.2 (Obtain Ω) For each flow assignment in Step 1.1, find the maximal feasible $X = (x_1, x_2, \dots, x_n)$ satisfying both constraints (11) & (12):

$$\begin{cases} \alpha_1 f_{rj} + \alpha_2 h_{rj} \leq x_{rj} \leq \alpha_1 f_{rj} + \alpha_2 h_{rj} + (\alpha_1 - 1) & \text{for } j = 1, 2, \dots, n_r, \\ x_i = M_i & \text{for } a_i \notin K_r \end{cases} \quad (11)$$

$$\sum_{i=1}^n c_i x_i \leq B. \quad (12)$$

The capacity x_{rj} is derived according to (11) to guarantee that the MC K_r supports at most (d_1, d_2) , and to meet the budget (12).

APPENDIX

A. Proof of Theorem 1

Proof: (i) Suppose to the contrary that $(x_{rj} - \alpha_1 f_{rj} - \alpha_2 h_{rj}) \geq 1$ for a_{rj} . The flow assignment $(f'_{r1}, f'_{r2}, \dots, f'_{rn_r}, h'_{r1}, h'_{r2}, \dots, h'_{rn_r}) = (f_{r1}, f_{r2}, \dots, f_{rj} + 1, \dots, f_{rn_r}, h_{r1}, h_{r2}, \dots, h_{rn_r})$ is feasible under X because $\alpha_1 f'_{rj} + \alpha_2 h'_{rj} = (\alpha_1 f_{rj} + \alpha_2 h_{rj} + 1) \leq x_{rj}$, and $\sum_{j=1}^{n_r} f'_{rj} = d_1 + 1$ & $\sum_{j=1}^{n_r} h'_{rj} = d_2$. This contradicts that X is a (d_1, d_2, B) -MC. (ii) If $x_i \leq M_i - 1$ for an $a_i \notin K_r$, then $X_1 = (x_1, x_2, \dots, x_i + 1, \dots, x_n)$ with $X_1 > X$ supports at most (d_1, d_2) . This is a contradiction.

B. Proof of Theorem 2

Proof: Suppose that X is a (d_1, d_2, B) -MC, but $X \notin \Omega_{\max}$. Lemma 2 shows that $X \in \Omega$. Then there exists a $Y \in \Omega_{\max}$ such that $Y > X$ (Y supports at most (d_1, d_2)), which contradicts that X is a (d_1, d_2, B) -MC. Hence, $X \in \Omega_{\max}$.

Conversely, suppose that $X \in \Omega_{\max}$ but it is not a (d_1, d_2, B) -MC. Then $V(X) = (d_1, d_2)$, and there exists a (d_1, d_2, B) -MC Y such that $Y > X$. Lemma 2 says that $Y \in \Omega$, which contradicts that $X \in \Omega_{\max}$. Hence, X is a (d_1, d_2, B) -MC.

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